

1-Assignment Sec Arch & Design

2/28/2026

85/100 Points

Attempt 1

Review Feedback
2/24/2026Attempt 1 Score:
85/100View Feedback²Anonymous Grading: **no****Unlimited Attempts Allowed**

2/14/2026 to 3/5/2026

▼ **Details**

Assignment Objective

Students will

- design a **secure architecture for a realistic application**,
- perform **formal threat modeling**,
- identify security risks across **application, infrastructure, and organizational layers**, and
- propose **defensible security controls** with justification.

Assignment Design Checklist

- Choose a **realistic application scenario**
- Define **assets, trust boundaries, and actors**
- Perform structured **threat modeling**
- Design **security architecture controls**
- Justify decisions with **risk-based reasoning**
- Produce a **clear, professional report**

Scenario (Choose One)

Students must choose **one** scenario. No inventing fantasy systems.

Option A. Online Payment Processing Application

Includes:

- Web frontend
- API backend
- Payment gateway integration
- Admin portal

- User database
- Merchant Database
- Core Banking System

Option B. Healthcare Appointment System

Includes:

- Patient portal
- Doctor portal
- Hospital / Health Care System
- API backend
- Third-party integrations
- Sensitive personal data

Option C. Internal Enterprise Application

Includes:

- Employee access
- Role-based permissions
- Internal APIs
- Logging and monitoring
- Integration with identity provider
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Option D. University Management System

Includes:

- Students Access of
 - LMS,
 - Attendance Management System,
 - Fees Management System,
 - Grades,
 - Alumnis
 - Wellness
 - Library Management System
- Faculty access of
 - LMS
 - Attendance
 - Salary distribution
 - Their Courses
 - Library
 - Wellness
- Administration / Registrar
- Role-based permissions
- Internal APIs
- Logging and monitoring
- Integration with identity provider
 - May be Job Portals

- Integration with Companies so that their jobs could appear in University Portal for upcoming graduating Students

Option E. A Human living in Karachi could be threat modeled

Which could range from

- His movements
- his material belongings
- his spiritual beliefs
- how can he be attacked and robbed
- how can one be provided security
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Option F. Any organization could be threat modeled

in terms of

- their assets
 - Hardware
 - Software
 - People
- The CIA aspects of their assets, information, secrets, business plans, etc.

Option F. Gym Membership

Which could range from

- Club Memberships
- Trainers
 - their qualifications
 - their salaries
- member
 - their profiles
 - their goals
 - their assignments to trainers
 - their fees
- Gym Upkeep
 - monthly maintainance
 - Administrators
- Spotify access
 - it's playlist
 - it's play time
 - so on and so forth

Threat Model Readings.

<https://medium.com/appsec-untangled/threat-modeling-handbook-4-how-to-get-better-at-threat-modeling-699e4cb336d5> (<https://medium.com/appsec-untangled/threat-modeling-handbook-4-how-to-get-better-at-threat-modeling-699e4cb336d5>)

Constraints

- Cloud-agnostic architecture
 - No vendor-specific services required
 - Must assume **internet-facing exposure**
 - Must consider **insider and external threats**
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Task 1. System Definition and Architecture

Student Actions

- Define:
 - Application components
 - Users and roles
 - Data types handled
 - External dependencies
- Draw a **high-level architecture diagram**
- Identify **trust boundaries**

Required Elements

- Frontend
- Backend services
- Data storage
- External integrations
- Admin access paths

Evidence Required

- Architecture diagram
 - Written description of components and boundaries
-

Task 2. Asset Identification and Security Objectives

Student Actions

- Identify **critical assets**, such as:
 - Credentials
 - Personal data
 - Financial data
 - Business logic

- Define **security objectives**:
 - Confidentiality
 - Integrity
 - Availability
 - Accountability

Evidence Required

- Asset inventory table
 - Mapping of assets to security objectives
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Task 3. Threat Modeling

Students must perform **structured threat modeling** using **one** of the following:

- STRIDE
- OWASP Threat Modeling

Required Threat Areas

- Authentication
- Authorization
- Data storage
- API communication
- Logging and monitoring
- Administrative access

For Each Threat

- Describe the threat
- Identify affected component
- Explain impact
- Assign risk level (Low/Medium/High)

Evidence Required

- Threat model table
 - Threat diagram or annotated architecture
 - Risk reasoning for each threat
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Task 4. Secure Architecture Design

Students must propose **architectural security controls**, not code fixes.

Required Control Categories

- Identity and access management
- Network segmentation
- Data protection
- Secrets management
- Monitoring and logging
- Secure deployment practices

Examples:

- API gateways and auth layers
- Trust boundary enforcement
- Separation of admin and user planes
- Defense-in-depth controls

Evidence Required

- Updated architecture diagram
 - Written justification for each control
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Task 5. Risk Treatment and Residual Risk

Student Actions

- Decide how to handle each high-risk threat:
 - Mitigate
 - Transfer
 - Accept
 - Avoid
- Explain **why** some risks remain

Evidence Required

- Risk treatment table
 - Residual risk explanation
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Task 6. Final Architecture and Threat Report

Report Must Include

- 1. System overview
- 2. Architecture diagram
- 3. Asset inventory
- 4. Threat model
- 5. Security controls
- 6. Residual risks
- 7. Assumptions and limitations

Submission Guidelines

- Submit everything in Github Profile / Github Repo.

View Rubric

Select Grader

Salman Khwaja Muhammad

Some Rubric		
Criteria	Comments	Points
Architecture Design view longer description	Comment Encryption at rest is being done on which algo ?	15/20 pts
Asset & Risk Identification view longer description	Comment Why all Risk Levels, HIGH... ?	10/15 pts
Threat Modeling Quality view longer description	Comment The idea of threat model is not make diagram. It's to keep the diagram a living breathing document so that it encapsulates the risk associated with it.	25/30 pts
Secure Architecture Controls view longer description	Comment	20/20 pts
Documentation & Clarity	Comment	15/15 pts

Criteria	Comments	Points
view longer description		

<https://github.com/breehaqasim/Cybersecurity-Theory-Tools/tree/main/Secure%20Architecture%20and%20Design> 
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(<https://hulms.instructure.com/courses/4979/modules/items/187030>)



(<https://hulms.instructure.com/courses/4979/modu>)

You are unable to submit to this assignment as your enrollment in this course has been concluded.

